

Resolving the Fuzzy Dark Matter Tension via Environment-Dependent Scalar Fields: PTA Signatures and Bayesian Inference Framework

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Fuzzy Dark Matter (FDM) provides a compelling resolution to the small-scale crises of Cold Dark Matter (CDM). However, it is currently hindered by a severe environmental mass-spectrum tension: dwarf galaxy cores require an ultralight boson ($m_a \sim 10^{-23}$ eV) to support macroscopic quantum pressure, whereas the kinematic coherence of cold stellar streams (e.g., GD-1) and Lyman- α forest data strictly bound the mass to $m_a \gtrsim 10^{-21}$ eV. We resolve this tension by embedding FDM within a scalar-tensor Effective Field Theory (EFT) featuring a conformal coupling to the baryonic density. This interaction dynamically shifts the effective mass of the scalar field, remaining light in diffuse dwarf spheroidal halos while becoming heavy in the baryon-dense disk of the Milky Way. Furthermore, we derive the exact General Relativistic spatial correlation kernel (Overlap Reduction Function) for this massive scalar field, demonstrating a strictly monopolar timing residual signature. Finally, we establish a rigorous Bayesian framework using the ENTERPRISE pulsar timing array software to statistically isolate this scalar dark matter signature from the stochastic Hellings-Downs background of supermassive black hole binaries.

I. INTRODUCTION

The cold dark matter (CDM) paradigm is highly successful at describing the large-scale structure of the universe. However, at galactic scales, it encounters persistent phenomenological challenges, most notably the “core-cusp” problem [1]. Fuzzy Dark Matter (FDM), modeled as an ultra-light scalar field or axion, offers an elegant resolution. For a boson of mass $m_a \sim 10^{-23}$ eV, the macroscopic de Broglie wavelength reaches ~ 1 kpc, generating a quantum pressure tensor that smooths out central densities into flat soliton cores [2].

Despite this success, FDM is currently under severe observational tension. To successfully reproduce the flat-topped density profiles of dwarf galaxies like IC 2574, the scalar mass must be strictly bounded near 10^{-23} eV. Conversely, an axion this light induces intense gravitational potential fluctuations (“granulation”) over large scales. Phase-space analyses of cold stellar streams in the Milky Way, such as the GD-1 stream, demonstrate that such fluctuations would dynamically heat and rapidly disperse these structures. Preserving stellar streams demands a much heavier particle, $m_a \gtrsim 10^{-21}$ eV [3].

In this paper, we resolve this mass tension by promoting the FDM axion to an environmentally dependent scalar field. We construct an Effective Field Theory (EFT) wherein the scalar mass dynamically scales with the local background matter density. We then derive the rigorous General Relativistic observational signatures of this scalar field in Pulsar Timing Arrays (PTAs) and establish a Bayesian framework to detect it.

II. DENSITY-COUPLED EFT FOR FUZZY DARK MATTER

We construct an EFT where the ultra-light scalar field ϕ is non-minimally coupled to the stress-energy tensor of standard matter via a conformal factor. The action in the Einstein frame is given by:

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V_0(\phi) + \mathcal{L}_m(\tilde{g}_{\mu\nu}, \psi_m) \right], \quad (1)$$

where M_{Pl} is the reduced Planck mass, and the Jordan frame metric $\tilde{g}_{\mu\nu}$ is related to the Einstein frame metric by $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$.

Assuming a non-relativistic pressureless dust background for standard matter ($T \approx -\rho_b$), the coupling induces an effective potential for the scalar field:

$$V_{\text{eff}}(\phi) = V_0(\phi) + [A(\phi) - 1]\rho_b. \quad (2)$$

Following the standard Damour-Polyakov conformal expansion [4], we set $A(\phi) \approx 1 + \frac{\beta}{2M_{\text{Pl}}^2} \phi^2$. The bare potential of the ultra-light field is modeled as $V_0(\phi) = \frac{1}{2} m_0^2 \phi^2$.

The effective mass of the scalar field is defined by the second derivative of the effective potential evaluated at its minimum:

$$m_{\text{eff}}^2 = \left. \frac{\partial^2 V_{\text{eff}}}{\partial \phi^2} \right|_{\phi_{\text{min}}} = m_0^2 + \frac{\beta \rho_b}{M_{\text{Pl}}^2}. \quad (3)$$

Resolution of the FDM Tension: Equation (3) provides an elegant resolution to the observational conflict. In the dark matter halos of dwarf spheroidal galaxies, the central baryonic density is exceptionally low ($\rho_b \rightarrow 0$). The field relaxes to its bare mass $m_0 \sim 10^{-23}$ eV, preserving the kiloparsec-scale de Broglie wavelength necessary for quantum core formation.

Conversely, within the disk of the Milky Way where cold stellar streams orbit, the local baryonic overdensity is orders of magnitude higher. The conformal coupling term $\frac{\beta \rho_b}{M_{\text{Pl}}^2}$ dominates the potential, dynamically shifting the effective mass to $m_{\text{eff}} \gtrsim 10^{-21}$ eV. The scalar field behaves locally as a heavier, colder particle. This drastic reduction in the de Broglie wavelength suppresses the granular potential fluctuations, preserving the phase-space coherence of the GD-1 stream.

III. GR DERIVATION OF THE PTA SPATIAL KERNEL

To empirically test this model in the Milky Way halo environment ($m_{\text{eff}} \sim 10^{-21}$ eV $\rightarrow f \sim 10^{-7}$ Hz), we must compute the exact metric perturbations induced by the massive scalar field and their resulting impact on pulsar timing residuals.

Following Khmelnitsky and Rubakov [5], an oscillating scalar dark matter background $\phi = \phi_0 \cos(m_a t)$ induces an oscillating pressure $p_\phi \simeq -\rho_\phi \cos(2m_a t)$. In the linearized Newtonian gauge, the perturbed metric is:

$$ds^2 = -(1 + 2\Phi)dt^2 + (1 - 2\Psi)\delta_{ij}dx^i dx^j. \quad (4)$$

Solving the linearized Einstein field equations yields a purely isotropic, time-dependent gravitational “breathing mode”:

$$\Phi(t) = \Psi(t) = \frac{\pi G \rho_\phi}{m_a^2} \sin(2m_a t). \quad (5)$$

The timing residual $R(t)$ for a pulsar located at a spatial distance L in the direction of the unit vector $\hat{p}$ is the integral of the fractional frequency shift $z(t) = \delta\nu/\nu$:

$$z(t) = \frac{1}{2} \frac{\hat{p}^i \hat{p}^j}{1 - \hat{n} \cdot \hat{p}} \partial_t h_{ij}. \quad (6)$$

Because the metric perturbation is an isotropic scalar field ($h_{ij} \propto \delta_{ij}$), the geometric projection factor simplifies strictly to 1. The scalar field effectively modulates the clock rate at Earth, independently of the pulsar’s sky location.

The Overlap Reduction Function (ORF), $\Gamma(\theta_{ab})$, defines the cross-correlation of timing residuals between two pulsars a and b separated by an angle θ_{ab} on the sky:

$$\Gamma(\theta_{ab}) = \frac{\langle R_a(t) R_b(t) \rangle}{\sqrt{\langle R_a^2(t) \rangle \langle R_b^2(t) \rangle}}. \quad (7)$$

Assuming the “pulsar terms” (the metric perturbation at the distant pulsars) are decoherent over kiloparsec distances, the cross-correlation isolates the “Earth term”—the local metric oscillation at the Solar System Barycenter (SSB). Because this scalar breathing mode acts uniformly on all photon geodesics, the dominant signature is a pure **monopole**:

$$\Gamma_{\text{Scalar}}(\theta_{ab}) = 1.0. \quad (8)$$

Taking into account the relative velocity of the Earth through the FDM halo, a velocity-suppressed dipole correction emerges. However, this rigorously derived monopole signature guarantees a positive-definite covariance matrix, establishing a definitive prediction for scalar FDM in PTA data. This stands in orthogonal contrast to the spin-2 quadrupolar Hellings-Downs correlation of supermassive black hole binaries.

IV. BAYESIAN MCMC INFERENCE FRAMEWORK

To validate the detectability of this model, we move beyond deterministic FFT simulations and establish a comprehensive Bayesian inference pipeline utilizing the ENTERPRISE (Empirical Network for the Terrestrial Extraction of Residuals in Pulsars by Robust Inference) suite [6].

A. Likelihood and Covariance Formulation

The timing residuals for a network of N pulsars are modeled as $\delta\vec{t} = M\vec{\epsilon} + F\vec{a} + \vec{n} + \vec{s}_{\text{GWB}} + \vec{s}_{\text{FDM}}$, containing timing ephemeris adjustments (M), red noise (F), uncorrelated white noise ($\vec{n}$), the spin-2 GWB, and the scalar FDM signal.

The likelihood function $\mathcal{L}$ is a multivariate Gaussian:

$$\mathcal{L}(\delta\vec{t}|\vec{\lambda}) = \frac{1}{\sqrt{\det(2\pi\mathbf{C})}} \exp\left(-\frac{1}{2}\delta\vec{t}^T \mathbf{C}^{-1} \delta\vec{t}\right), \quad (9)$$

The total covariance matrix $\mathbf{C}$ incorporates the specific spatial kernels:

$$\mathbf{C}_{ab} = \mathbf{C}_{\text{noise}}\delta_{ab} + \Gamma_{\text{HD}}(\theta_{ab})\mathbf{C}_{\text{GWB}} + \Gamma_{\text{Scalar}}(\theta_{ab})\mathbf{C}_{\text{FDM}}. \quad (10)$$

B. MCMC Implementation

Using Parallel Tempering Markov Chain Monte Carlo (PTMCMCSampler), we sample the posterior probability distribution $P(\vec{\lambda}|\delta\vec{t})$. We define the spatial correlation as an orthogonal basis function within ENTERPRISE:

```

1 import enterprise.signals.parameter as parameter
2 from enterprise.signals import signal_base,
  gp_signals, utils
3 from enterprise.signals import
  spatial_correlations as sc
4 import numpy as np
5
6 # 1. Define the Scalar-Tensor Monopole Kernel
7 @sc.spatial_interp
8 def scalar_breathing_mode(theta):
9     """Rigorous GR-derived ST Kernel: Gamma =
10     1.0 (Earth-term Monopole)."""
11     return np.ones_like(theta)

```

```

12 # 2. Setup Priors for the FDM Signal (10^-23 to
    10^-21 eV)
13 log10_A_fdm = parameter.Uniform(-18, -12)
14 log10_f_fdm = parameter.Uniform(-9, -7)
15
16 # 3. Construct the FDM Gaussian Process Signal
17 fdm_plaw = utils.powerlaw(log10_A=log10_A_fdm,
    gamma=5.0)
18 fdm_signal = gp_signals.FourierBasisGP(
19     spectrum=fdm_plaw,
20     components=30,
21     spatial_corr=scalar_breathing_mode
22 )
23
24 # 4. Model Assembly (Adding HD GWB + FDM
    Monopole)
25 # model = tm + wn + rn + gwb_hd + fdm_signal

```

Listing 1. ENTERPRISE framework definition for scalar FDM isolation.

To statistically isolate the scalar FDM signal from the SMBHB background, we utilize the Savage-Dickey density ratio to compute the Bayes Factor ($\mathcal{B}$) between competing hypotheses ($\mathcal{M}_{\text{HD}}$ vs $\mathcal{M}_{\text{HD}+\text{Scalar}}$). The algorithm prevents cross-contamination between the tensor and scalar signals. A Bayes factor of $\ln \mathcal{B} > 3$ constitutes

strong evidence for the scalar field.

V. CONCLUSION

We have demonstrated that modeling Fuzzy Dark Matter as a density-coupled Effective Field Theory provides a dynamic, physical resolution to the tension between dwarf galaxy cores and cold stellar streams. By deriving the exact scalar-tensor metric perturbations from first-principles General Relativity, we have shown that this field imprints a strict monopolar correlation on Pulsar Timing Array residuals.

By embedding this theoretical prediction within the rigorous ENTERPRISE Bayesian inference framework, we transition the theory from analytical speculation to a strictly falsifiable, statistically robust hypothesis ready to be tested against the latest NANOGrav and EPTA datasets.

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